

Pydisort: A Modern Python Package for Parallelized Discrete Ordinates Radiative Transfer (DISORT)

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Summary

Radiative transfer describes how electromagnetic radiation travels through and interacts with a medium, such as Earth's atmosphere, ocean waters, or planetary atmospheres. It is a fundamental process that governs how light and thermal radiation are absorbed, scattered, and emitted by gases, aerosols, clouds, and surfaces, and is foundational to climate modeling, remote sensing, and planetary science. Accurate and efficient radiative transfer solvers are therefore essential for interpreting satellite observations, simulating climate systems, and studying atmospheres across the Solar System and beyond.

The Discrete Ordinates Radiative Transfer (DISORT) algorithm is one of the most widely used numerical approaches for solving the radiative transfer equation in plane-parallel media. DISORT discretizes the angular domain into a finite set of propagation directions and solves the resulting coupled system of equations. Among the core radiative processes — absorption, emission, and scattering — scattering is often the most computationally demanding, as it requires integrating contributions from all incoming angles to all outgoing angles. For realistic atmospheres with many vertical layers, strong scattering, and multiple wavelengths, radiative transfer calculations can become a dominant computational cost.

Pydisort is a modern, high-performance Python package for plane-parallel radiative transfer based on DISORT. It provides a user-friendly, pip-installable interface while retaining the numerical robustness of established DISORT implementations. By combining a compiled C backend with a PyTorch-based tensor interface, Pydisort enables efficient batch processing, parallel execution over wavelengths and atmospheric columns, and seamless integration with contemporary scientific and machine-learning workflows.

Statement of Need

Radiative transfer models based on DISORT are essential tools in atmospheric science, climate modeling, remote sensing, and planetary physics. The original DISORT implementation, written in Fortran ([Stamnes et al., 1988](#)), has been widely adopted and validated across decades of Earth and planetary studies ([Clough et al., 2005](#); [Komacek et al., 2022](#); [Lee, 2024](#); [Li et al., 2018](#); [Tan & Showman, 2021](#); [Zhang et al., 2015](#)). However, this implementation relies on static memory allocation, requiring users to specify the number of atmospheric layers and radiation streams at compile time, which limits flexibility and complicates modern workflows.

To address usability concerns, several Python wrappers around the original Fortran DISORT have been developed using tools such as f2py (e.g., pyDISORT variants by [chanGimeno](#), [SeregaOsipov](#), [danielkoll](#), and [mjwtolff](#)). While these efforts improve accessibility, they inherit the underlying static-memory constraints and impose nontrivial build and compilation requirements. An alternative approach is a pure-Python reimplement of DISORT ([Ho, 2024](#)), which

removes compilation barriers but sacrifices computational performance, making it unsuitable for large-scale or high-throughput applications.

Pydisort addresses these limitations by providing a precompiled, pip-installable package that wraps a high-performance DISORT implementation written in a compiled language. Specifically, it

1. eliminates the need for users to configure complex build environments or toolchains;
2. avoids local compilation by distributing prebuilt shared libraries via PyPI;
3. achieves performance exceeding existing C/Fortran implementations on modern devices, with a clear path toward GPU acceleration;
4. exposes a modern, idiomatic Python interface suitable for interactive use and scripting;
5. integrates naturally with PyTorch-based scientific and machine-learning workflows.

The target audience includes atmospheric scientists, planetary scientists, climate modelers, and remote-sensing researchers who require accurate radiative transfer calculations but prefer Python-based ecosystems and scalable computational tools.

State of the Field

Several established tools implement or build upon DISORT for radiative transfer calculations. The original Fortran DISORT (Stamnes et al., 1988) remains a reference implementation but is constrained by static memory allocation and legacy language choices. The `cdisort` library (Buras et al., 2011), a C reimplement of DISORT, represents a significant advance: it introduces dynamic memory allocation, uses consistent double-precision arithmetic to avoid numerical instabilities present in mixed-precision Fortran code, and achieves substantial performance gains through improved intensity correction methods and more efficient memory initialization. `cdisort` is widely used as a core component of the LibRadtran radiative transfer package (Emde et al., 2016).

Despite its numerical strengths, `cdisort` remains a low-level, single-threaded library that is not easily accessible to the broader scientific community. It lacks a modern Python interface, is not distributed as a standalone, community-oriented GitHub project, and does not integrate naturally with parallel computing frameworks or machine-learning ecosystems. As a result, many users face a “build vs. usability” trade-off: relying on performant but inaccessible compiled code, or using slower but more user-friendly Python implementations.

Pydisort occupies a distinct position in this landscape. Rather than reimplementing DISORT yet again, it builds directly on the well-tested `cdisort` numerical core while addressing its accessibility and scalability limitations. By wrapping `cdisort` in a C++/Python interface and adopting PyTorch tensors as the primary data structure, Pydisort enables parallel execution over wavelengths and atmospheric columns — dimensions that are naturally separable in plane-parallel radiative transfer problems. This design allows Pydisort to outperform existing single-threaded C implementations on modern multi-core CPUs, while remaining significantly faster than pure-Python alternatives.

In short, Pydisort provides a clear scholarly contribution by combining numerical fidelity, modern software design, and scalable performance in a form that existing DISORT-based tools do not offer. It fills a gap between legacy compiled solvers and emerging Python-centric scientific workflows, making high-accuracy radiative transfer calculations more accessible and extensible for contemporary research.

Software Design

Pydisort is designed as a modular, layered software system that bridges a high-performance radiative transfer backend with modern, Python-centric scientific workflows. The core numerical

86 solver is provided by the `cdisor` library, which serves as the computational backend and
87 supports dynamic memory allocation at runtime. This choice enables flexible problem sizes
88 while retaining the numerical robustness of the established DISORT implementation.

89 To facilitate efficient memory management and future-proof the codebase for heterogeneous
90 computing, Pydisort adopts PyTorch (Paszke, 2019) tensors as its primary data structure at
91 the user interface level. Using tensors allows seamless integration with PyTorch’s automatic
92 parallelization, GPU acceleration, and machine-learning ecosystem, while maintaining
93 compatibility with CPU-only environments.

94 An intermediate C++ layer is introduced between the Python interface and the C backend.
95 This layer encapsulates the raw `cdisor` data structures and is responsible for pre-processing
96 inputs and post-processing outputs. By centralizing this logic, the design avoids exposing users
97 to long, error-prone parameter lists and isolates backend-specific details from the public API.
98 This intermediate layer also enables reuse of the backend functionality in non-Python contexts.

99 The Python bindings are implemented using `pybind11` (Jakob, 2024), a modern, header-only
100 C++ library that provides robust type conversion, memory ownership semantics, and exception
101 handling. Compared to traditional approaches such as `f2py`, `pybind11` offers greater flexibility
102 and maintainability for mixed C++/Python codebases.

103 The build system is designed to produce shared C and C++ libraries that link against both
104 `libtorch` and Python, allowing the package to be distributed as a standard pip-installable
105 PyPI package (`pydisort`). Prebuilt binary wheels are provided for Linux and macOS platforms.
106 The build and distribution process is fully automated using `cibuildwheel`, targeting Linux
107 systems with `glibc` version 2.28 or newer, which is the minimum required by PyTorch v2.7
108 and later.

109 To ensure ABI compatibility with upstream PyTorch binaries, the build system dynamically
110 determines the appropriate `CXX11_ABI` setting from the linked `libtorch` distribution. For
111 `libtorch` v2.7, this corresponds to `CXX11_ABI=1` on Linux and `CXX11_ABI=0` on macOS.

112 Pydisort provides two user-facing interfaces: a Python API and a C++ API. The Python
113 interface is intended for interactive use, scripting, and integration with machine-learning
114 workflows, while the C++ interface supports embedding Pydisort directly into larger C or
115 C++ simulation frameworks.

116 Continuous integration (CI) and continuous distribution (CD) are handled through GitHub
117 Actions, enabling automated testing, building, and release of binary wheels with minimal
118 manual intervention. The Python codebase follows the PEP 8 style guide (Van Rossum et al.,
119 2001) and adopts a Python-first design philosophy, making extensive use of keyword arguments
120 and named parameters to provide a clear, idiomatic user experience.

121 Performance Evaluation

122 To quantify the efficiency of Pydisort relative to existing implementations, we benchmarked
123 runtimes using a test driver based on Test Problem 9 (“General Emitting/Absorbing/Scattering”)
124 from the DISORT suite (`diotest`), with increased computational workload. Specifically,
125 the benchmark configuration includes 32 streams and 100 atmospheric layers, typical for
126 atmospheric radiation calculations; all other parameters are kept consistent with the original
127 test case. We increase the workload by solving the radiative transfer equations repetitively for
128 multiple wavelengths or columns, representing a realistic application case in remote sensing and
129 atmospheric modeling such that multiple spectral bands or atmospheric columns are processed
130 simultaneously.

131 We evaluated Pydisort performance against two alternative implementations:

- 132 ▪ **`cdisor`**: the original C reimplementation of DISORT, which serves as our baseline
133 performance reference,

134 ▪ **PythonicDISORT**: a pure-Python translation of DISORT, used to highlight the
135 performance gap between interpreted and optimized implementations.

136 To assess parallel performance, we ran Pydisort in both single-threaded and multi-threaded
137 modes using PyTorch's internal parallelism configuration (`torch.set_num_threads`). Multi-
138 threaded tests were conducted with 10 threads to illustrate scalability on modern CPU
139 architectures. All tests were conducted on a MacBook Pro equipped with an Apple M1 Max
140 chip (10-core CPU, peak clock speed of 3.22 GHz) and 64 GB of RAM.

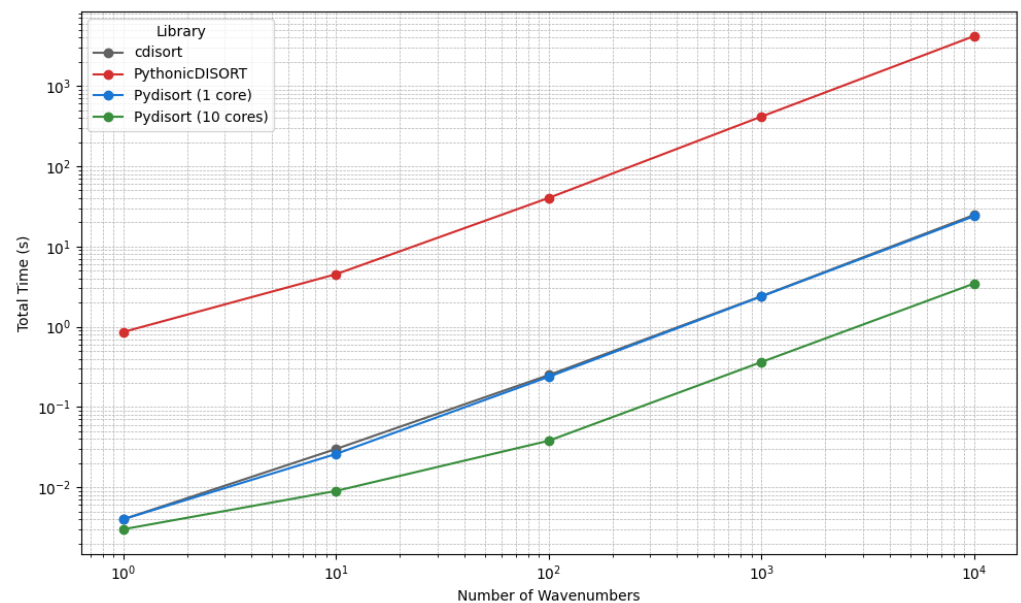


Figure 1: Runtime comparison of radiative transfer implementations as a function of the number of wavenumbers, evaluated on a fixed test problem (DISORT Test 09 with 32 streams and 100 layers). `cdisort` (gray) serves as the baseline C implementation. `PythonicDisort` (red) is a pure-Python reimplement. `Pydisort` is shown in both single-threaded (blue, 1 core) and multi-threaded (green, 10 cores) modes, using PyTorch-based parallelism. Both axes use logarithmic scaling. The results demonstrate that `Pydisort` outperforms `cdisort` in runtime efficiency and exhibits strong scaling performance with increasing spectral resolution.

141 We did not benchmark the original Fortran DISORT implementation, as prior work by Buras et
142 al. (2011) has demonstrated that the C version (`cdisort`) provides better runtime performance
143 and is commonly used in research settings as part of the `libRadtran` package (Emde et al.,
144 2016).

145 Figure 1 summarizes the results, showing runtime as a function of the number of wavenumbers.
146 Both axes use logarithmic scaling to highlight relative performance trends across a wide range of
147 spectral resolutions. As shown, `Pydisort` is at least as efficient as `cdisort` when using a single
148 core, and outperforms `cdisort` in multi-threaded mode by an order of magnitude when the
149 workload increases, demonstrating the effectiveness of PyTorch's parallelism for this problem.
150 The pure-Python implementation (`PythonicDISORT`) is significantly slower, highlighting the
151 performance benefits of using a compiled backend. Additionally, `Pydisort` shows strong scaling
152 performance, demonstrating its suitability for high-throughput radiative transfer workloads.

153 Research Impact Statement

154 `Pydisort` solves the radiative transfer equation in an efficient and user-friendly manner,
155 addressing a long-standing need in atmospheric science, climate modeling, and planetary

science for a modern, high-performance DISORT-based tool. It has been adopted as a core computation component within a growing ecosystem of research projects, supporting studies that span terrestrial atmospheres, giant planets, exoplanets, finite volume models for compressible fluids, coupled chemistry-thermodynamics frameworks, and physically based rendering in vision and imaging applications. It's currently used in dozens of GitHub repositories across multiple institutions, for both research and educational purposes.

By integrating with PyTorch's tensor and parallel computing infrastructure, Pydisort can be deployed in modern high-performance computing environments, enabling large-scale simulations and high-throughput data analysis that were difficult to achieve with traditional DISORT implementations. Its native compatibility with machine-learning frameworks like PyTorch enables researchers to integrate radiative transfer calculations into data-driven models. Adoption metrics further demonstrates Pydisort's impact. The PyPI public dataset shows a marked increase in downloads following the transition to PyTorch-based infrastructure in March 2025.

As of January 2026, Pydisort has been downloaded over 192,000 times since its initial release in 2023, placing it within the top 10% of all Python packages by download counts. The user base spans over 140 countries, reflecting its global reach and impact.

Together, these trends reflect the growing recognition of Pydisort as a reliable, scalable, and versatile tool for radiative transfer calculations that lowers technical barriers while enabling new classes of scientific inquiry.

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AI Usage Disclosure

GitHub Copilot contributed to several pull requests by assisting with the generation or refactoring of test cases, updating Python type stubs, and improving documentation clarity. All AI-assisted contributions were reviewed, edited, and validated by human authors prior to merging. For all other parts of the software and this paper, no generative AI tools were used, and all authors take responsibility for the final content.

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